Exercise 4

1. Hypotheses

- **Null hypothesis (H_0):** $\mu_B = \mu_A$ (or $\mu_B \leq \mu_A$, meaning Model B does not improve true prediction accuracy compared to Model A).
- **Alternative hypothesis (H_1):** $\mu_B > \mu_A$ (meaning Model B has a higher true prediction accuracy than Model A).

2. Decision at $\alpha = 0.05$

Yes, the data scientist should claim that Model B is significantly better than Model A.

Explanation: Since the p-value (0.04) is less than the significance level $\alpha = 0.05$ ($0.04 < 0.05$), we reject the null hypothesis H_0 in favor of the alternative hypothesis H_1. This indicates that observing an improvement at least this large by random chance alone is unlikely under H_0, providing statistically significant evidence at the 5% level that Model B improves accuracy.

3. Decision at $\alpha = 0.01$

If the significance level were set to $\alpha = 0.01$, the decision would change: we would **fail to reject** H_0.

Explanation: Because the p-value (0.04) is greater than the stricter significance level $\alpha = 0.01$ ($0.04 > 0.01$), the test result is no longer statistically significant at the 1% level. There would not be sufficient statistical evidence at the 1% significance level to claim that Model B performs better than Model A.

Exercise 5

Problem 1

Problematic statement: "Because the observed average loading time was lower after optimization, we selected a one-sided test."
Issue: The choice between a one-sided and a two-sided test must be made *before* looking at the sample data, based on the research question or prior claim. Selecting a one-sided test after observing the direction of the sample data (post-hoc) invalidates the test's Type I error rate.
Correction: "Because our goal is to test the web engineer's prior claim that the optimization reduces average page-loading time, we specified a one-sided test before collecting and examining the data."

Problem 2

Problematic statement: "The p-value of 0.07 means that there is a 7% probability that the optimization has no effect."
Issue: The p-value is not the probability that the null hypothesis (H_0) is true. Instead, it is the probability of observing a sample result at least as extreme as the one obtained, *assuming* that the null hypothesis (H_0) is true.
Correction: "The p-value of 0.07 means that if the optimization truly had no effect on average

page-loading time (H_0 were true), there would be a 7% probability of observing a sample reduction in average loading time at least as large as the one we observed."

Problem 3

Problematic statement: "Because $p > 0.05$, we accept H_0 and conclude that the optimization does not improve loading performance."
Issue: In hypothesis testing, we never "accept H_0" or conclude that H_0 is proven true. A p-value greater than α only means that we lack sufficient evidence to reject H_0.
Correction: "Because $p = 0.07 > 0.05$, we fail to reject H_0 at the 5% significance level. There is insufficient statistical evidence at the 5% level to conclude that the optimization reduces average page-loading time."

Problem 4

Problematic statement: "Therefore, it should not be deployed."
Issue: A decision should not be based solely on whether a p-value crosses an arbitrary $\alpha = 0.05$ cutoff (especially with a borderline p-value of 0.07). It ignores practical significance (effect size and confidence interval), the statistical power of the test, and the cost-benefit trade-off of deployment.
Correction: "While the reduction is not statistically significant at the 5% level ($p = 0.07$), the decision to deploy should also consider the estimated effect size (confidence interval), the statistical power of the study, and the engineering cost versus potential performance benefit."